

Two questions

Why the two–gluon rows carry no ultraviolet divergence — and which diagrams give DGLAP with a fragmentation function

$$S \equiv p^+ + k^+, \quad \zeta \equiv \frac{k^+}{S}, \quad r \equiv y' - y$$

Short answers. (1) Your field–theory instinct is right, and it is not violated: these rows are *cut* diagrams, not loops. The unobserved gluon is **on shell**, integrated over three–dimensional phase space, not over $\int d^4\ell$. That does not by itself guarantee finiteness — real–emission integrals can still diverge at large $p_\perp$ — but here they do not, for a structural reason given in §2. And the ultraviolet divergence that field theory demands **is** present in this calculation: it sits in the **virtual** row, $\mathfrak{A}^{(1)} \times \mathfrak{A}^{(3)}$, as exactly $-\frac{2b_0}{\epsilon}$ (§3). **(2)** Fragmentation DGLAP comes from a splitting that happens **after** the last interaction with the target, and it has three unmistakable kinematic signatures (§4). **The two– $\mathfrak{C}$ row has all three** — which means my earlier description of it as an initial–state, PDF–type collinear pole was wrong. It is a fragmentation contribution (§5).

1. A cut is not a loop

In the cross section the unobserved gluon appears on both sides of the cut, at the same momentum. Its integration is

$$\int \frac{dp^+ d^2p_\perp}{2p^+ (2\pi)^3} \quad (\text{on shell, } p^- = \frac{p_\perp^2}{2p^+} \text{ fixed}),$$

not $\int \frac{d^4\ell}{(2\pi)^4}$ over an off–shell line. Ultraviolet divergences in field theory come from the deep off–shell region of a genuine loop; a cut line has no such region — one integration is removed by the mass–shell delta. So the naive counting that gives Λ_{UV}^2 or $\log \Lambda_{UV}$ for a loop simply does not apply to these rows.

But that is not the whole answer, and it would be sloppy to stop there. A real–emission integral *can* still diverge at large $p_\perp$ if the matrix element does not fall fast enough — that is exactly what a collinear–factorisation counterterm sometimes has to remove. In transverse position space, large $p_\perp$ is the region where points collapse, which is precisely what every scan in this work has been testing. So the finiteness is a *result*, not an automatic consequence. §2 says why it comes out that way.

2. Why the collapse regions converge

The rule established throughout: in two transverse dimensions a logarithmic short–distance divergence needs the integrand to go as ρ^{-2} , because n collapsing points supply $\rho^{2(n-1)} \frac{d\rho}{\rho}$. A single Weizsäcker–Williams kernel is only ρ^{-1} . **To get ρ^{-2} you need two kernels singular on the same separation.**

Now look at where the unobserved gluon sits. Its transverse position z is **shared** between amplitude and conjugate — it is one integration variable, not two. The two kernels attached to it join z to **different** points:

$$\frac{(y-z)^m}{(y-z)^2} \quad (\text{amplitude}), \quad \frac{(y'-z)^{m'}}{(y'-z)^2} \quad (\text{conjugate}),$$

and y and y' are held apart by the measurement phase $e^{-ik \cdot (y' - y)}$. To make both kernels singular at once you must collapse y , y' and z together — three points, measure $\rho^4 \frac{d\rho}{\rho}$, against an integrand of only ρ^{-2} . The measure wins by ρ^2 .

The one place a genuine ρ^{-2} does appear is a vanishing light–cone energy denominator, e.g. $D = p^+(x-z)^2 + k^+(x-w)^2 \rightarrow 0$ when x, z, w collapse. But that too needs three points: integrand ρ^{-2} against $\rho^3 d\rho$. Checked explicitly in the $|\mathfrak{B}_2|^2$ and $\mathfrak{B}_3$ rows — convergent, with room to spare.

3. The ultraviolet divergence is where it should be

Every row examined in this work, sorted by whether the unobserved gluon is cut or virtual:

row	character	transverse UV?	what it does have
two $\mathfrak{C}$ vertices	real (cut)	no (26 regions)	IR log at $ z \rightarrow \infty$, coefficient $P_{gg}(\zeta)$
$ \mathfrak{B}_2 ^2$, two ρ	real	no (26)	IR log at $ z \rightarrow \infty$
$ \mathfrak{B}_2 ^2$, three ρ	real	no (57)	IR log at $ z \rightarrow \infty$, coefficient $-\frac{1}{2p^+k^+} \times WW \times WW$
$\mathfrak{B}_3$ row	real	no (26)	IR log in term 3; a $\frac{1}{\Lambda}$ power divergence at $p^+ = k^+$; an on–contour pole
$\mathfrak{A}\bar{\mathfrak{C}} \times \mathfrak{B}_2$	real	no (26)	no IR either — the phase protects z
$\mathfrak{A}^{\dagger(1)} \times \mathfrak{A}^{(3)}$	virtual	YES	$-\frac{2}{\epsilon} \cdot \frac{11}{3} N_c = -\frac{2b_0}{\epsilon}$, plus $-\frac{67}{9} + \frac{\pi^2}{3}$

That last line is the point. $\mathfrak{A}^{(3)}$ is the NLO correction to the *single*–gluon emission amplitude — a genuine virtual loop — and it carries a genuine ultraviolet pole whose coefficient is exactly $b_0 = \frac{11N_c}{3}$, i.e. the one that renormalises g^2 . The real rows have no UV; the virtual row has the coupling–renormalisation UV. **That is the standard structure of any NLO calculation**, and finding it is a check that the rows are being counted correctly, not an anomaly.

4. What a fragmentation contribution looks like

Both DGLAP flavours use the same P_{gg} ; what distinguishes them is *when* the splitting happens relative to the target, and that shows up in three places.

	initial state $\rightarrow$ PDF	final state $\rightarrow$ fragmentation
(A) diagram	parent splits <i>before</i> the shockwave; both daughters cross separately, so Wilson lines appear at both daughter positions , $U(y)U(z)$	parent crosses the shockwave and splits <i>after</i> ; the Wilson lines see only the parent's position , and the daughters enter only through the splitting kernel
(B) collinear region	daughter collinear to the beam : small $k_\perp$ of a daughter	the two final–state gluons collinear to each other , $M^2 = \frac{(p^+k_\perp - k^+p_\perp)^2}{p^+k^+} \rightarrow 0$, i.e. relative $q_\perp \rightarrow 0$ — which in position space is large relative separation , not a collapse
(C) kinematics	hard factor at the measured k ; the convolution rides on the incoming flux	hard factor at the parent momentum k/ξ , with $E \frac{d\sigma^h}{d^3k} = \int \frac{d\xi}{\xi^2} D(\xi) \left[E' \frac{d\hat{\sigma}}{d^3k'} \right]_{k'=k/\xi}$

So, concretely, the diagrams that give you a **gluon fragmentation function** are those in which a *single* gluon is produced from the source, picks up its Wilson line, and *then* undergoes a $g \rightarrow gg$ splitting with one daughter measured and one integrated out — the splitting vertex sitting downstream of every U . Diagrams where the splitting precedes the shockwave, so that both daughters scatter independently, give the initial–state (PDF–type) log instead.

A *quark* fragmentation function would need a $g \rightarrow q\bar{q}$ vertex, i.e. P_{qg} rather than P_{gg} . None of the vertices in this calculation ($\mathfrak{A}$, $\mathfrak{B}_2$, $\mathfrak{B}_3$, $\mathfrak{C}$) are quark vertices, so at this order you can only generate the gluon FF.

5. The two– $\mathfrak{C}$ row is a fragmentation contribution

A correction to what I told you earlier. In the DGLAP note I wrote that $r = y' - y$ is conjugate to the measured momentum k , and therefore that the $\frac{1}{\epsilon}$ should be absorbed into the *gluon distribution*. The first statement is wrong, and the second follows from it. Here is the correct kinematics.

The two $\mathfrak{C}$ deltas gave $w - z = \frac{S}{k^+}(y - z)$ and $w' - z = \frac{S}{k^+}(y' - z)$, so

$$w' - w = \frac{S}{k^+}(y' - y) = \frac{r}{\zeta} \quad \implies \quad e^{-ik \cdot (w' - w)} = e^{-i\left(\frac{k}{\zeta}\right) \cdot r}.$$

r is conjugate to $\frac{k_\perp}{\zeta}$, **not** to $k_\perp$. And $\frac{k_\perp}{\zeta}$ is the **parent's** transverse momentum: in a collinear splitting the daughter takes the fraction ζ of both the longitudinal *and* the transverse momentum, so $k_\perp^{\text{parent}} = k_\perp/\zeta$. The hard factor is being evaluated at k/ζ — criterion (C).

All three signatures are then present:

1. **Coefficient** $P_{gg}(\zeta)$ with $\zeta = \frac{k^+}{p^+ + k^+}$, the fraction of the *parent* carried by the *measured* gluon — the fragmentation variable.
2. **Measure** $\frac{d\zeta}{\zeta^2}$, which is exactly the $\frac{dz}{z^2}$ of the collinear fragmentation formula for the invariant cross section — not the $\frac{d\xi}{\xi}$ of a parton–density convolution.
3. **Hard factor** at k/ζ , from the phase above.

And criterion (B) fits too: the divergence of that row sits at $|z| \rightarrow \infty$ — large relative separation, i.e. small relative $q_\perp$, the two final–state gluons becoming collinear to each other. It is *not* at any collapse of points, which is why it never looked ultraviolet.

So the pole should be absorbed into a **gluon fragmentation function** $D_{g/g}(\zeta, \mu_F)$ by the counterterm $-\frac{1}{\epsilon} \frac{P_{gg}}{2} \otimes (\text{LO})$ on the *outgoing* leg, with the hard factor kept at k/ζ .

5.1 The one check that would settle it independently

The three signatures above are kinematic and unambiguous. The structural confirmation is criterion (A): in the reduced colour structure the unobserved gluon's Wilson line **cancelled** ($U^{d'b}(z)U^{db}(z) = \delta^{d'd}$), leaving one Wilson line per side, at y and y' — the parent positions — with z surviving only inside the splitting kernel. That is the final–state pattern of (A). To make it airtight rather than suggestive, check the time ordering directly in the derivation of the $\mathfrak{C}$ vertex: whether its $\delta^{(2)}$ places the splitting before or after the shockwave is fixed there, and it is the one ingredient I cannot verify from the expressions alone.

zint/src/note_frag.py (§5), and the row results already in RESULTS_row8...row15 (§3).